

Panagiotis Liampas

64 Bay State Rd, Boston, MA 02215 | (857) 799-0500 | pliam005@mit.edu

[linkedin.com/in/panagiotis-liampas](https://www.linkedin.com/in/panagiotis-liampas) | github.com/pliam1105 | pliam1105.github.io | medium.com/@pliam1105

Education

Massachusetts Institute of Technology, Cambridge, MA

May 2028

Candidate for **Bachelor of Science in Artificial Intelligence and Decision Making**

GPA: 5.0/5.0

Relevant Coursework: Advances in Computer Vision, Robotics: Science and Systems (<https://rss2025-15.github.io/website/>), Robotic Manipulation, Deep Learning, Representation Inference and Reasoning in AI, Introduction to Machine Learning, Dynamical System Modeling and Control Design, Electrical Circuits: Modeling and Design of Physical Systems, Linear Algebra, Introduction to Computer Science Programming in Python, Mathematics for Computer Science

Mandoulides High School, Thessaloniki Greece

May 2023

High School Diploma

SAT: 1550, GPA: 19.8/20.0

Work Experience

Epsilon Orosimo Software, Thessaloniki, Greece

September 2023 - October 2023

Junior Software Developer

- Converted and developed custom extension modules for the **Pylon ERP software, using C#**, from another software, enhancing the capabilities and integrating custom operations with regards to the **UI, logic and backend**.

Alpha Systems, Thessaloniki, Greece

August 2022 - October 2022

Embedded Systems Intern

- Implemented an **IoT weather station**, which is connected using LoRaWAN to the **TTN Network and the Tago.io platform** and presents the data in a dashboard.
- Cooperated with the **Nokia technical team** to integrate it with the **Nokia IOC platform**.

Research Experience

MIT Sensing, Perception, Autonomy, and Robot Kinetics Lab, Cambridge, MA

September 2025 – December 2025

Undergraduate Research Opportunity

- Designed a **visual-only approach** that uses a single camera, on board a robot, to **reconstruct a 3D scene in real time** using **Simultaneous Localization And Mapping**, and create a map with the **right granularity to perform complex tasks**, achieving **efficient task-driven open-set semantic segmentation** of the scene, and improving upon current approaches on both the **usefulness** of the resulting map and the **real-time performance** of the algorithm.

MIT Autonomous Underwater Vehicle Lab,

June 2025 – August 2025

Cambridge, MA

Undergraduate Research Opportunity

- Designed a modular and generalizable **sensor fusion system** and **Extended Kalman Filter based state estimator**, for the Remote Explorer (REx) IV vehicle, combining **camera, LiDAR, radar, GPS, and IMU data** of REX along with radio-transmitted information of the contact vehicles, to compute useful information like the **position and velocity** of REX and the other vehicles/obstacles, as well as the **size** of those.

MIT Laboratory for Atomistic and Molecular Mechanics, Cambridge, MA

February 2025 – May 2025

Undergraduate Research Opportunity

- Designed a new bio-inspired generative design framework, synthesizing the emergent dynamics of spacetime-inhomogeneous cellular automata, and the pattern-finding capabilities of transformer neural network (TNN) models, to be able to explore a larger geometric configuration space to further optimize the mechanical properties of architected materials.

BlueDot Impact AI Safety Fundamentals Alignment Course, Remote May 2024 – June 2024
Independent Research [Certificate](#), [GitHub](#), [Article Part 1](#), [Article Part 2](#)

- Conducted a review of various state-of-the-art **Distributional Reinforcement Learning** methods, as well as **risk measures** that can be integrated into the optimization operators of **Q-Learning**, using knowledge of the **distribution of the possible rewards** at a specific state of the agent.
- Evaluated their **effectiveness in decreasing the risk and ensuring the safety of self-driving cars** at worst-case scenarios, by **training a Machine Learning model** in a **simulated highway environment** and capturing the results.

Non-Trivial Fellowship, Remote July 2023 - October 2023
Independent Research [Preprint](#)

- Designed “Risk-averse Batch Active Inverse Reward Design”, an approach improving upon “Active Inverse Reward Design” in terms of **safety and robustness of AI models** and **reducing uncertain, potentially dangerous behaviors** after deployment, using **Reinforcement Learning from Human Feedback**. Specifically, I implemented a process that finds the intended reward function by asking queries to the human, and repeating that process on real-world data, allowing its adaptability to new environments and improving its efficiency. It also takes the actions that are the most certain, increasing the safety and control over the process.

Dr. Kevrekidis (Johns Hopkins), Remote July 2023 - October 2023
Research Collaborator [GitHub](#)

- Trained a Neural Network** (using TensorFlow) to **predict the behavior** of the Reduced MSP Model, from the **dynamical system** described in the paper: “On the parameter combinations that matter and on those that do not”, based on data derived from time integration of the system with various parameters and start states using the Runge-Kutta method.

Dr. Koumoutsakos (Harvard), Remote August 2023 - October 2023
Research Collaborator [GitHub](#)

- Implemented a **simulated environment for Braitenberg vehicles**, to examine their interactions with each other, sources of various types, and certain “constrained” areas where the vehicles cannot be in.

Awards & Accomplishments

Greek Mathematics Team

Silver Medal (1 place from Gold), International Mathematical Olympiad, Chiba, Japan	July 2023
Silver Medal, International Mathematical Olympiad, Oslo, Norway	July 2022
Silver Medal (1 place from Gold), Balkan Mathematical Olympiad, Antalya, Turkey	July 2022
Silver Medal, Balkan Mathematical Olympiad, Agros, Cyprus	July 2022

Greek Astronomy Team

Bronze Medal, International Olympiad in Astronomy and Astrophysics, remote	November 2021
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Greek Informatics Team

Bronze Medal, Balkan Olympiad in Informatics, Maribor, Slovenia	October 2023
Participation, International Olympiad in Informatics	2021, 2022, 2023
Honorable Mention (2 places from Bronze), Balkan Olympiad in Informatics, remote	October 2023

Non-Trivial Fellowship July 2023 – September 2023
Third Prize (4th – 8th place), for designing “Risk-Averse Batch Active Inverse Reward Design”

Build Teams

MIT Arcturus Autonomous Robotics , Autonomy subteam	September 2024 – present
Autonomy co-lead	April 2025 – present
Captain	March 2026 – present
- Leading the team (~30 people) as the team captain starting March 2026, setting a long-term plan and individual short-term goals for the team and coordinating with the subteam leads to ensure their timely completion, as well as managing the team's finances, sponsorships, fundraising, and recruitment. - Co-leading the Autonomy subteam starting April 2025, coordinating a team of ~12 people and being responsible for assigning tasks to the subteam members and ensuring their timely completion to high standards, with the goal of completing the competition's tasks using robust foundational subsystems. - Implemented a ROS node that computes a sequence of buoy pairs, as well as a preferable region (center & radius) in between them, that the autonomous boat should go through to follow a specific path, based on an obstacle map that's provided as an input. - Improved the fusion of LiDAR and Camera data in the autonomous catamaran, using PCL and OpenCV, so that the resulting point clouds and image segments best depict the detected objects, increasing the accuracy and robustness of the perception system of the robot on detecting buoys and banners (using RANSAC) on the water. - Implemented an Extended Kalman Filter based Simultaneous Localization And Mapping (SLAM) algorithm that uses the refined distance, angle, and color detections of the buoys and the GPS-based position and IMU-calculated orientation and speed of the robot, to create an accurate map of the buoys' and banners' positions and the robot's pose, and capture the uncertainty of those predictions, increasing their accuracy over time. - Enhanced our perception system by incorporating two additional cameras facing to the sides and backwards, merging the buoy detections from all three cameras and thus increasing the FoV of our robot, improving the robustness of the perception system, also leading to better localization and mapping, when performing tasks. - Ported the EKF SLAM pipeline to a factor-graph implementation using GTSAM, implementing both the odometry measurements and object detections as corresponding factors and using an outlier-robust cost function to improve the robustness of the resulting map.	

Personal Projects

SLAM Bot	November 2023 - February 2024
Individual project	Article , GitHub
Engineered a 3D printed robot, including its chassis, moving mechanisms, and electronics, that uses the Robot Operating System (ROS 2) and Simultaneous Localization and Mapping (SLAM) algorithms to map, locate itself, and navigate around an unknown area.	
Rover Hand	December 2022 - February 2023
Individual project	GitHub
Programmed an autonomous robot that can detect objects, track, and catch them with its robotic hand, using computer vision and machine learning.	
Parking Detection	August 2022 - January 2023
Individual project	GitHub , Municipality Website
Implemented a system which uses machine learning to automatically detect empty parking lots and highlight them on a website, which was given for use by the municipality and the local community to solve the problem of the difficulty finding parking spaces.	
Plant Watering	July 2022 – November 2022

Individual project

[GitHub](#)

- A system that **measures the humidity of the soil under some plants** and **sends the data to a website** to be used to **determine the amount of irrigation needed**.

Rover Sensors

July 2018 – August 2019

Individual project

[GitHub](#)

- Implemented a **remotely controlled prototype of a rover** with **integrated sensors** that can potentially be used to determine the habitability of a planet and **show the data, the results and the rover's route in a website, an android app and a windows app**.

Programs/Courses

The Knowledge Society Global Innovate, Remote

September 2023 – June 2024

Completion

- Studied a broad range of areas and recent developments in various fields.
- Acquired a comprehensive understanding of the methods used to solve global problems, from examining their root causes to refining a concrete idea and plan for their mitigation, and applied them in internal challenges and hackathons.
- Conducted a review wrote an article about the fundamentals of Simultaneous Localization And Planning. [Article](#)
- Built a robot that applies the knowledge I acquired from studying the field of autonomous navigation.
- Conducted a review and wrote an article about state-of-the-art methods of LiDAR-Camera fusion, their characteristics and their limitations. [Article](#)

Hellenic Institute of Advanced Studies AI Summer School 2024,

July 2024

NCSR Demokritos, Athens

Attendance

- Gained a comprehensive understanding of the fundamentals and state-of-the-art methods of Machine Learning, as well as their applications in a wide range of areas and ethical implications.

MITx's Circuits and Electronics XSeries Program, Remote

July 2024

Circuits and Electronics 1: Basic Circuit Analysis

Circuits and Electronics 2: Amplification, Speed, and Delay

[Certificate](#)

- Gained a comprehensive understanding of fundamental circuit concepts, analysis techniques, design methods, and applications in various real-world devices, acquiring the tools to be able to apply those in diverse situations and use cases.

Online Publications

A review of LiDAR-Camera fusion for robust perception in complex environments

July 2024

Medium

[Article](#)

- Examining how Autonomous Mobile Robots navigate in unstructured environments, using multiple sensors to improve robustness and accuracy and mitigate collisions and false positives.

Risk-Sensitive Reinforcement Learning with Applications in Autonomous Driving

June 2024

Medium

[Part 1](#), [Part 2](#)

- An exploration of how Distributional Reinforcement Learning and appropriate risk measures can be used to improve safety, evaluated in a simple highway environment.

Building a 3D-printed robot that uses SLAM for autonomous navigation

January 2024

Medium

[Article](#)

- A thorough step-by-step writeup of the design and building process of a 3D-printed autonomous robot that uses Jetson Nano, Arduino, and the Nav2 package in Robot Operating System (ROS) 2.

Using SLAM to Autonomously Navigate in Unknown Environments

October 2023

Medium

[Article](#)

- A review of Simultaneous Localization and Mapping, consisting of an understanding of its sensor/hardware and algorithmic/software aspects, its applications, but also limitations and anticipated developments in the field.

Teaching Experience

Mandoulides High School , Thessaloniki, Greece	December 2022 - March 2024
Olympiad Preparation Teaching Assistant	
• Prepared students at Mandoulides High School for Mathematics and Informatics Olympiads.	
Mandoulides High School , Thessaloniki, Greece	November 2022 - February 2023
Robotics Mentor	
• Mentored the robotics team of Mandoulides High School for their participation in the Greek FIRST LEGO League.	

Leadership & Service

MIT Hellenic Student Association , Cambridge, MA	May 2025 – now
Officer	
• Planning and organizing events for the Greek community of MIT and the greater Boston area. • Helping in making the events successful and mitigating any issues that arise.	
Greenpeace Greece , Thessaloniki, Greece	September 2023 – August 2024
Project Coordinator	
• Organized initiatives to identify all the public water taps of Thessaloniki. • Organized and participated in awareness campaigns, increase the use of water taps by citizens to refill their bottles. • Increased the interest and capacity of the team by presenting the project to new members and facilitating their onboarding. • Got in touch with and convinced certain municipalities to fix the malfunctioning water taps, resulting in their improved state and increased use in many areas. • Developed the dashboard/map where the usable water taps are presented, including their location, status, and additional information.	
Arcturos Wildlife Shelter , Thessaloniki, Greece	September 2023 – August 2024
Volunteer	2 weeks, 8 hours/day
• Took care of ~20 Greek Shepherds, by feeding and walking them daily. • Contributed to various operations of the Bear Sanctuary and the Wolf Sanctuary.	

Skills & Knowledge

Language: English (Certificate of Proficiency in English, CEFR C2 level), French (Delf, CEFR B1 Level)

Technology

Building & programming **autonomous robots** and **electronics devices/embedded systems**, using **Arduino (C++) & Raspberry Pi (mostly Python, runs on Linux)** and **custom PCB** circuit boards designed with **Fritzing and EasyEDA**, including the integration of **sensors, motors, batteries/solar panels and custom power management circuits**, as well as **3D-printed mechanical parts & moving structures** designed using **Fusion 360**. Those are connected to an ecosystem of **websites and databases**, with which data is transmitted (**LoRaWAN, Wi-Fi, or GSM modules**) & processed using **JS, PHP, and SQL**, and accessed/used via a **web app/dashboard** using **HTML, CSS (& Bootstrap)**, or **Android (Android Studio & Java)** and **Windows UWP or .NET applications (Visual Studio & C#)**.

Knowledge in **Robotics, Localization, Mapping and Planning**, read “**Probabilistic Robotics**” by **Sebastian Thrun**, and practical experience in using the **Robot Operating System (ROS 2)**, specifically

publishing/subscribing to topics, working with various frames & transforms, implementing a **perception/localization/planning/navigation stack** using the respective packages and related nodes, using the **ROS 2 Control framework**, and designing drivers (hardware components) that use **Arduino for communication between the Raspberry Pi and the motors/servos**.

Integration of **Computer Vision and Machine Learning** technologies in autonomous systems and robots, using **Python or C++** based frameworks like **Point Cloud Library (PCL)** and **OpenCV**, for autonomous perception and sensor fusion, and **TensorFlow, Keras, and PyTorch** for object detection or Neural Network applications, theoretical and practical knowledge of **Q-Learning** and related decision-making algorithms used for determining the optimal policy for a **Markov Decision Process**, including the design, implementation, and evaluation of **risk-averse methods in the field of Reinforcement Learning from Human Feedback**, and ability to design and apply **inverse kinematics and control loop (PID)** mechanisms in an autonomous robot. Have also studied various methods used to fuse LiDAR (point cloud) and Camera (RGB/RGBD image) data, to improve accuracy and robustness of Autonomous Mobile Robots ([Article](#)).

Knowledge in **Dynamical Systems** and predicting their evolution, read “Nonlinear Dynamics And Chaos” by Steven H. Strogatz, as well as usage of **Neural Networks for fitting an ordinary differential equation**.

Knowledge in **Braitenberg Vehicles** and their properties, as well as their connection to human psychology, read “Vehicles: Experiments in Synthetic Psychology” by Valentino Braitenberg, and ability to **design physics simulations using Unity**.